



FUTURE SCHOLARS ETHIOPIA

PYTHON BASICS FOR BEGINNERS

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PART THREE





What will we learn?

- Operators
- Arithmetic Operators
- Comparison Operators
- Logical Operators
- Building Calculator-like program





What Are Operators?

- Operators are special symbols or keywords used to perform operations on values and variables.
 - e.g., $10 + 5$
- Here:
 - 10 and 5 are **operands**
 - $+$ is the **operator**
- Operators allow us to calculate, compare, and make decisions in our programs.





Arithmetic Operators

- Arithmetic operators are used for mathematical calculations.





Operator	Meaning	Example	Result
+	Addition	10 + 3	13
-	Subtraction	10 - 3	7
*	Multiplication	10 * 3	30
/	Division	10 / 3	3.333...
//	Floor division	10 // 3	3
%	Modulus	10 % 3	1
**	Exponentiation	2 ** 3	8





- Python follows an order when an expression contains multiple operators
 - e.g., $2 + 3 * 4 = 14$
- The multiplication happens first.
- Use parentheses when you want to control the order:
 - e.g., $(2 + 3) * 4 = 20$





Comparison Operators

- Comparison operators are used to compare two values.
- The result is always a Boolean value: **True** or **False**





Operator	Meaning	Example
==	Equal to	5 == 5
!=	Not equal to	5 != 3
>	Greater than	5 > 3
<	Less than	5 < 3
>=	Greater than or equal to	5 >= 5
<=	Less than or equal to	3 <= 5





Comparison Examples

age = 16

```
print(age == 16)
```

```
print(age > 18)
```

```
print(age < 18)
```

```
print(age != 15)
```

- Comparison operators are especially useful when making decisions with if statements.





= VS ==

- = means assignment.
 - e.g., age = 16
- == means comparison.
 - e.g., age == 16
- it asks “Is the value of age equal to 16?”
- Then the result will be True.





Logical Operators

- Logical operators are used to combine or modify conditions.
- Python has three main logical operators:
 - **and**
 - **or**
 - **not**
- They work with Boolean values.





and

- **and** returns **True** only when both conditions are true.

- e.g., age = 20
score = 90

```
if age > 18 and score > 50:  
    print("You have passed the  
        requirement")
```



or

- **or** returns **True** when at least one condition is true.

- e.g., **age = 20**
score = 40

```
if age > 18 or score > 40:  
    print("You can join the program")
```

not

- **not** reverses a Boolean value.
 - e.g., `answer = input("Do you have ID: ")`
`has_id = answer == "yes"`

`if not has_id:`
`print("You can join the program")`



Mini Project

- Build a calculator-like program that asks a user two numbers and displays their:
 - sum
 - difference
 - product
 - quotient and
 - remainder

